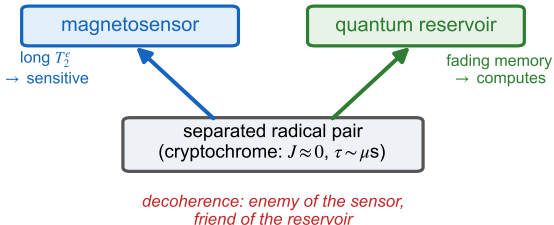
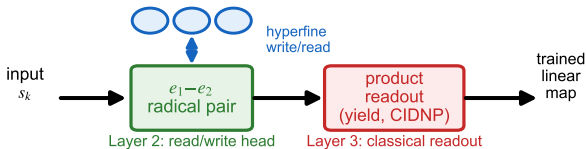


(a) one molecule, two functions



(b) nuclei are the register that must persist

Layer 1: nuclear-spin memory (long T_2)



$$\text{memory horizon} = \text{MC} \times T_{\text{turnover}}$$

reaches ms–s *if* the register is reused